Ordinary Differential Equations: Abridged Notes

Chris Ritter

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These abridged notes were reconstructed afterward because I did not take direct notes during class. They are not a record of the actual lecture.

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1 Topic Summary

Two solutions of a second-order homogeneous linear equation form a fundamental pair when they are linearly independent. Their Wronskian provides a test for independence. Abel's formula predicts how the Wronskian changes without requiring explicit solutions. These facts also summarize the existence and uniqueness ideas useful for review: a second-order linear initial-value problem has a unique solution on an interval where its normalized coefficients are continuous.

2 The Wronskian and Abel's Formula

For differentiable functions y_1, y_2, define

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2.$$

For two solutions of $y'' + p(t)y' + q(t)y = 0$, Abel's formula says

$$\boxed{W(t) = W(t_0) \exp\left(- \int_{t_0}^{t} p(s)\, ds \right).}$$

Thus on an interval where p, q are continuous, W is either nowhere zero or identically zero. A nonzero Wronskian means the pair is independent and spans all solutions of the homogeneous equation. When the leading coefficient vanishes, normalize only on intervals that do not cross that point.

2.1 Example: Compute a Wronskian Directly

Adapted from Boyce et al., 11th ed., §3.2, Problem 3. Find the Wronskian of $y_1 = e^{-2t}$ and $y_2 = te^{-2t}$.

Solution.

Since $y_1' = -2e^{-2t}$ and $y_2' = (1 - 2t)e^{-2t}$,

$$\begin{aligned} W &= e^{-2t}(1 - 2t)e^{-2t} - (-2e^{-2t})(te^{-2t}) \\ &= \boxed{e^{-4t}}. \end{aligned}$$

It never vanishes, so the two functions are linearly independent on all of $\mathbb{R}$.

2.2 Example: Determine a Wronskian Without Solving

Adapted from Boyce et al., 11th ed., §3.2, Problem 25. For two solutions of Legendre's equation

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0,$$

find the possible form of their Wronskian.

Solution.

Away from $x = \pm 1$, divide by $1 - x^2$. The coefficient of y' is $p(x) = -2x/(1 - x^2)$. Abel's formula gives

$$W(x) = C \exp\left(\int \frac{2x}{1 - x^2}\, dx \right) = C \exp\bigl(-\ln|1 - x^2|\bigr).$$

On any interval excluding -1 and 1, absorb the fixed sign of $1 - x^2$ into the constant to obtain

$$\boxed{W(x) = \frac{C}{1 - x^2}.}$$

If the solutions form a fundamental pair then $C \neq 0$. The three allowed maximal coefficient intervals are $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$.